

How Life Is Sustained on Earth

Air, water and sunlight work together — through photosynthesis and respiration — to keep life going.

The connections that keep life going

Earth has the right conditions for life, but it is the **connections between living and non-living things** that **sustain** it.

The oxygen–carbon dioxide cycle

- **Plants** use sunlight, carbon dioxide and water to make food by **photosynthesis**, releasing **oxygen**.
- **Animals and plants** use that oxygen for **respiration**, releasing carbon dioxide — which plants use again.

Sunlight, heat and water

- Sunlight powers photosynthesis and warms the Earth; the mild greenhouse effect keeps it warm enough for liquid water.
- **Water** covers about **70%** of Earth and is essential for all life.

Everything connects: air, water, soil, sunlight and living things interact — the same idea of interdependence you met in the ecosystem chapter.

Our responsibility: human activities can disturb these balances (e.g. raising CO₂ warms the planet), so we must protect Earth's delicate conditions.

Where students lose marks

Trap 1 — Photosynthesis releases oxygen; respiration uses it. Plants do both; the two processes balance the gases.

Trap 2 — A mild greenhouse effect is helpful — it keeps Earth warm; only too much is harmful.

Trap 3 — Life depends on links, not single factors. Air, water and sunlight work together to sustain life.

Model answers — write them like this

Example A. How do photosynthesis and respiration keep the air balanced?

Step 1: Plants take in CO_2 and release O_2 during photosynthesis.

Step 2: Living things use O_2 and release CO_2 during respiration.

$\therefore$ The two processes recycle oxygen and carbon dioxide, keeping the air fit for life.

Example B. Why is the greenhouse effect important for sustaining life?

Step 1: It traps some of the Sun's heat in the atmosphere.

Step 2: This keeps the temperature warm enough for water to stay liquid.

$\therefore$ A mild greenhouse effect keeps Earth warm enough for life.

[Ready to practise?](#) → [Open the worksheet](#)